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Dynamics of Glycolytic Oscillation Using Goldbeter- Lefever Reaction Model



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Abstract



This report investigates the complex non-linear dynamics and emergent oscillatory behavior of the glycolytic pathway, utilizing the classic two-variable Goldbeter-Lefever (GL) model. Glycolysis, a fundamental metabolic process, exhibits sustained periodic fluctuations under specific conditions, primarily driven by the allosteric regulation of the enzyme phosphofructokinase (PFK). By non-dimensionalizing the GL system, we isolate the core parameters governing this biochemical clock. A rigorous dynamical systems approach is employed, encompassing linear stability analysis, the calculation of the Jacobian matrix, and the identification of steady states. We demonstrate that the onset of glycolytic oscillations corresponds mathematically to a Hopf bifurcation, where the stable equilibrium point loses its stability and gives birth to a stable limit cycle. Furthermore, we leverage geometric singular perturbation theory to exploit the inherent time-scale separation between the rapid enzymatic turnover and the slower substrate accumulation. This fast-slow analysis reduces the high-dimensional bursting dynamics onto a lower-dimensional critical manifold, confirming the existence of relaxation oscillations. Numerical simulations using implicit integration schemes are presented to validate the theoretical phase portraits and bifurcation boundaries. Ultimately, this investigation bridges the gap between abstract dynamical systems theory and practical biochemical regulation, illustrating how cellular metabolic stability is maintained or destabilized by critical kinetic thresholds.

Glycolytic Oscillation

What is Glycolysis?

The fundamental metabolic pathway that cells use to extract energy from glucose, consisting of a sequence of ten enzyme-catalyzed reactions.

The Pacemaker Enzyme.

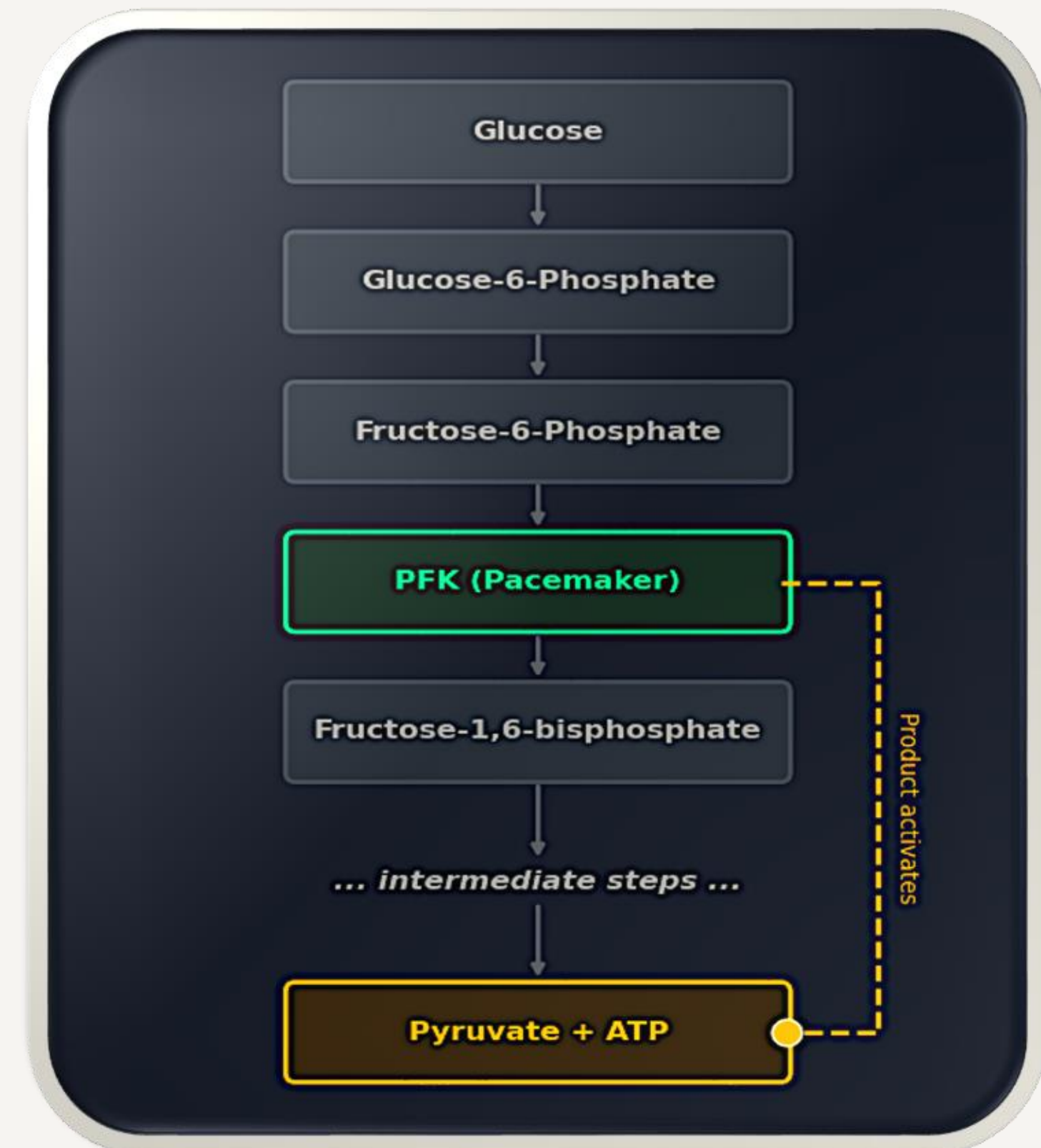
Phosphofructokinase (PFK) acts as the key regulatory enzyme that control the overall reaction rate through allosteric regulation.

The Phenomenon.

Under certain condition, metabolite concentration do not settle into a steady state. Instead, they exhibit sustained, periodic fluctuation – Glycolytic Oscillation.

Objective.

To mathematically model and analyze how these biological rhythms emerge using dynamical systems theory.



Goldbeter-Lefever Reaction Model

Allosteric Enzyme Kinetics

MWC Concerted Model

Based on the Monod-Wyman-Changeux (MWC) model of allosteric transitions, the GL model the cooperative binding and regulatory feedback that drive oscillatory behavior.

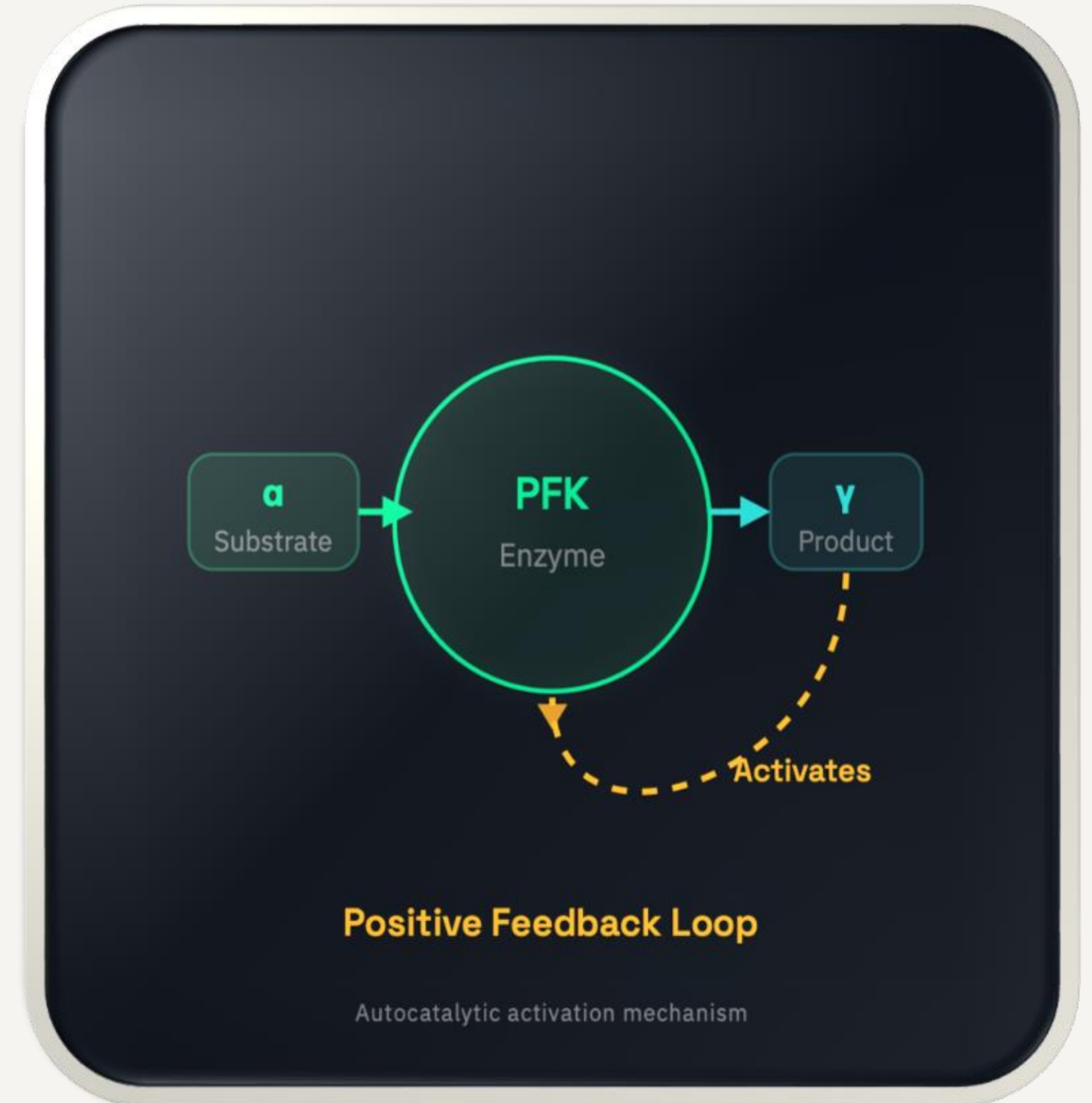
DIMENSIONLESS VARIABLES

α – Substrate concentration (Slow Variable)

γ – Product Concentration (Fast Variable)

GOVERNING EQUATIONS

$$\frac{d\alpha}{dt} = v - \sigma\Phi(\alpha, \gamma)$$
$$\frac{d\gamma}{dt} = q\sigma\Phi(\alpha, \gamma) - k_s\gamma$$



Stability Analysis & the Jacobian

JACOBIAN MATRIX

$$\begin{pmatrix} -\sigma \frac{\partial \Phi}{\partial \alpha} & -\sigma \frac{\partial \Phi}{\partial \gamma} \\ q\sigma \frac{\partial \Phi}{\partial \alpha} & q\sigma \frac{\partial \Phi}{\partial \gamma} \end{pmatrix}$$

CHARACTERISTIC EQUATION

$$\lambda^2 - \text{Tr}(J)\lambda + \det(J)$$

Trace: $\text{Tr}(J) = q\sigma\Phi_\gamma - \sigma\Phi_\alpha - k_s$

Determinant: $\det(J) = \sigma k_s \Phi_\alpha > 0$

Steady States

Found by setting $\frac{d\alpha}{dt} = 0$ and $\frac{d\gamma}{dt} = 0$ simultaneously.
The intersection of nullclines yields the equilibrium point(s)

Stability Determination

The sign of the trace and determinant of J at the equilibrium determine whether it is a stable node, stable spiral, or unstable spiral.

SINSTABILITY CONDITION $\text{Tr}(J) > 0$

Positive feedback overcomes stabilizing substrate consumption.

Hopf Bifurcation & Limit Cycles

Hopf Bifurcation

The mathematical boundary where a stable equilibrium point loses stability as a parameter crosses a critical threshold. Complex conjugate eigenvalues cross the imaginary axis from left to right

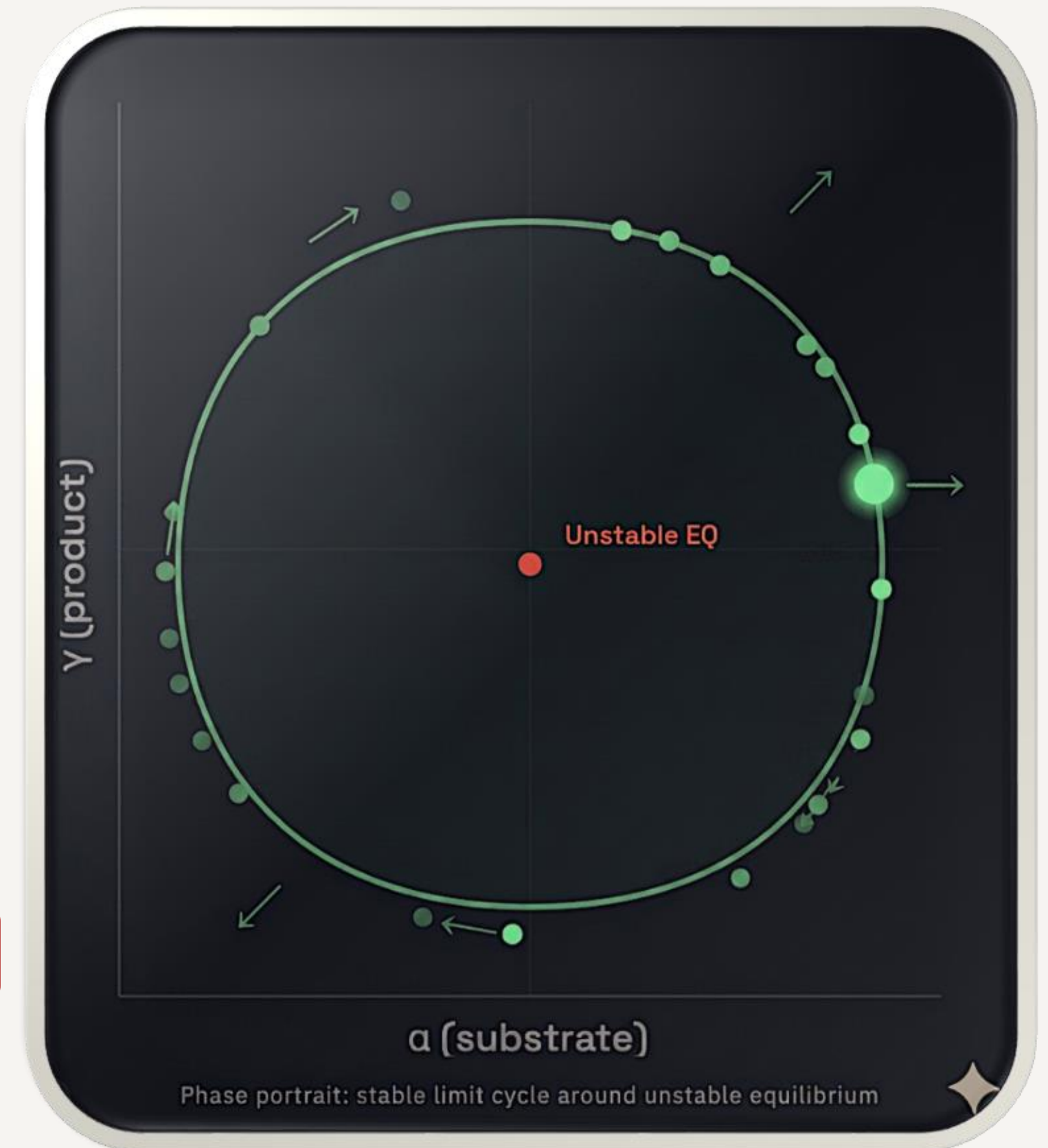
- At the bifurcation point: $Tr(J) = 0$ and $det(J) > 0$
- Eigenvalues: $\lambda = \pm \omega$ where $\omega = \sqrt{det(J)}$
- A pair of purely imaginary eigenvalues signals the onset of oscillation
- The GL model exhibits a **supercritical** Hopf bifurcation

Stable Limit Cycle

Periodic Orbit

Stable Limit Cycle

The Limit cycle represents the sustained, periodic oscillation observed experimentally
In yeast cells and muscle extracts



Fast-Slow Dynamics & the Critical Manifold

Time-Scale Separation

Substrate accumulation operation on a slow time scale $O(1)$, while enzyme turnover occurs on a fast time scale $O(\epsilon)$ where $\epsilon \ll 1$.

Critical Manifold M_0

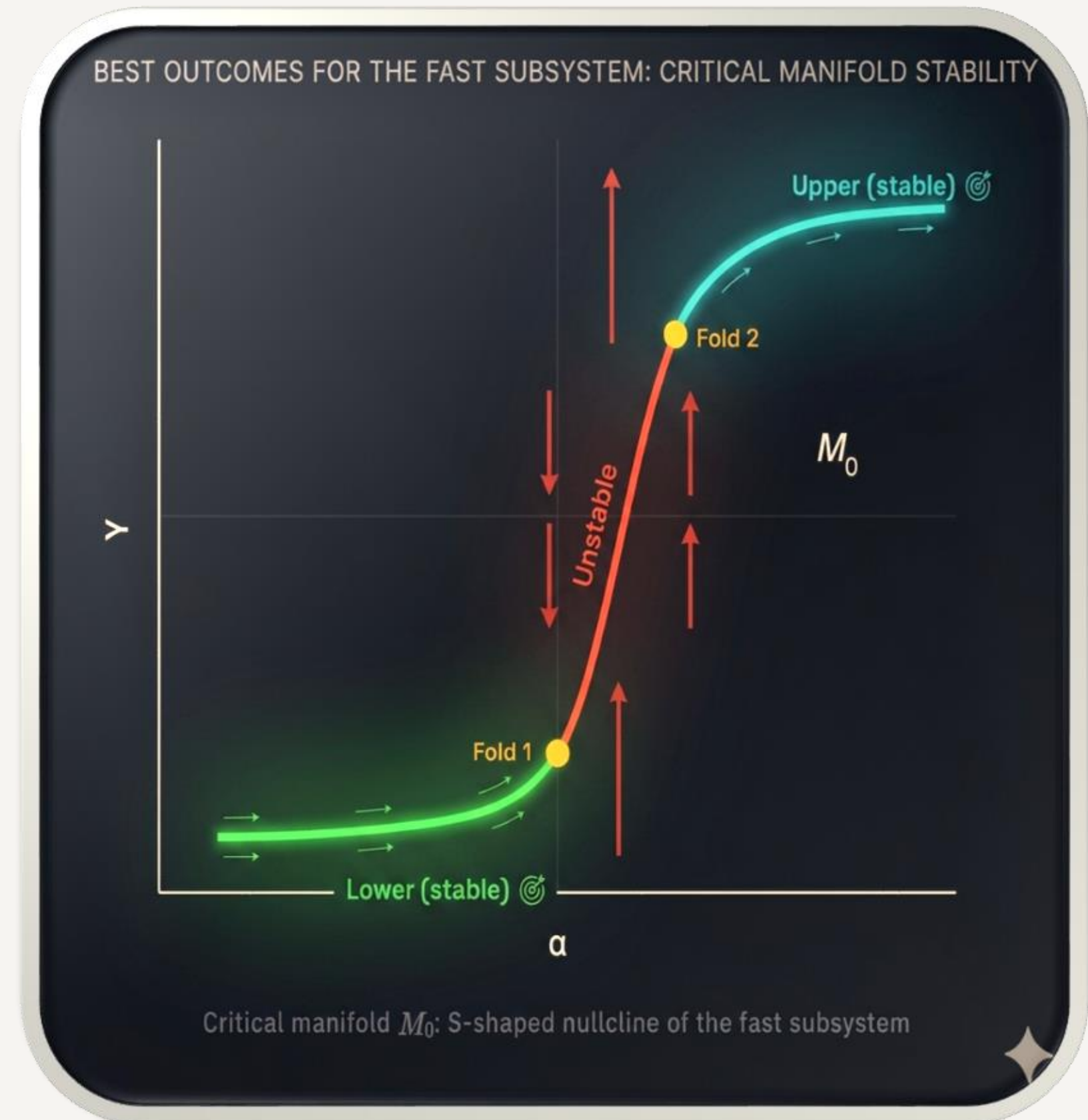
Taking the singular limit $\epsilon \rightarrow 0$ yields a cubic algebraic curve – the S-shaped nullcline that partitions phase space into distinct dynamic regimes.

-Lower branch: Enzyme off

-Middle branch: Unstable

- Upper branch: Enzyme saturated

The **fold points** (knees) of the S-curve are where the critical manifold normal hyperbolicity these are sites of rapid transitions between branches



Relaxation Oscillation

Geometry-Driven Oscillation

The trajectory follows the S-curve branches with slow evolution, punctuated by rapid jumps at the fold points – creating a characteristic relaxation cycle

→ **Charging**

Slow drift along the lower branch - substrate accumulates.

↑ **Firing**

Fast vertical jump at Fold 1 – enzyme activates explosively.

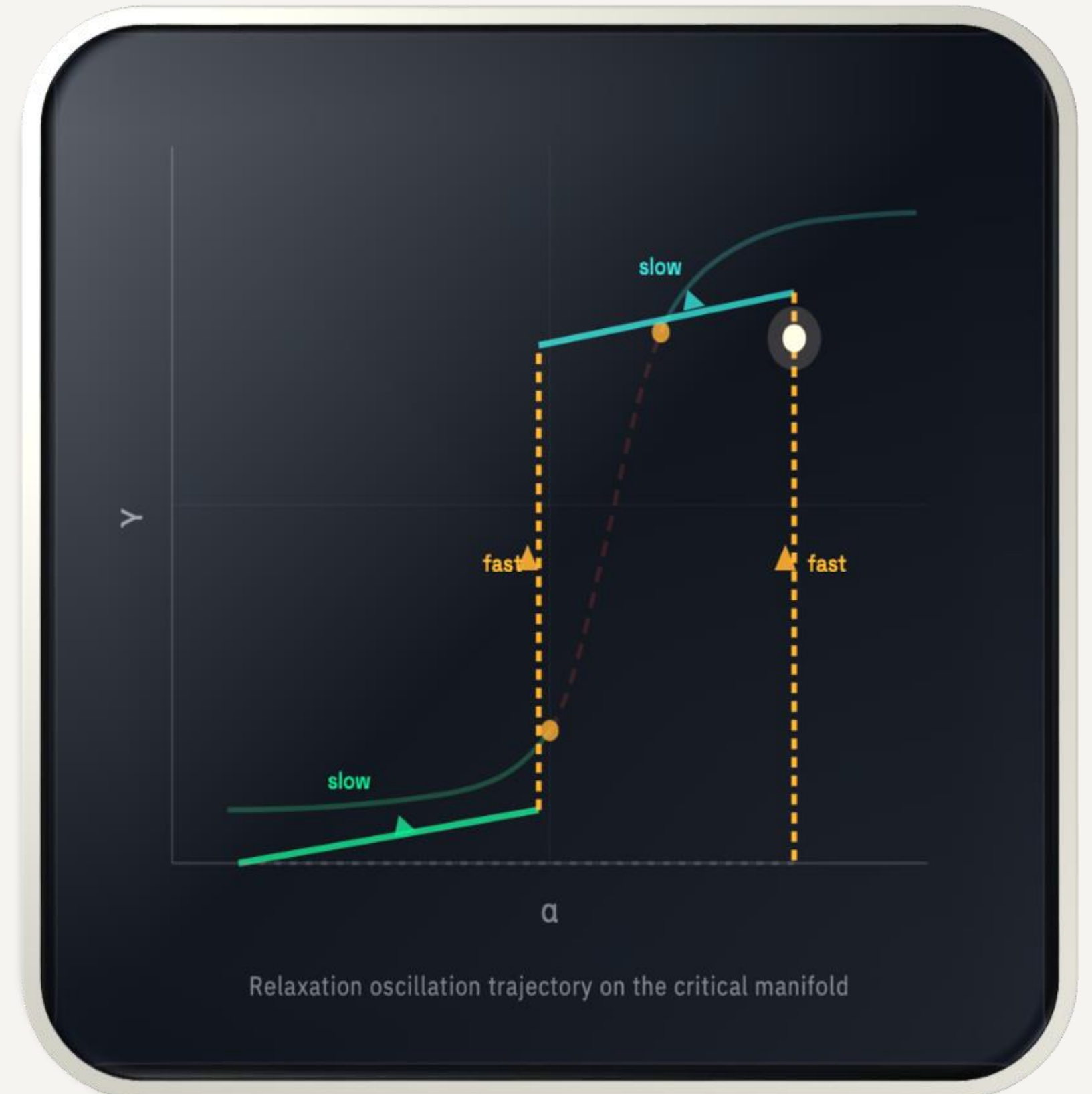
→ **Depleting**

Slow drift along the upper branch – substrate consumed.

↓ **Resetting**

Fast vertical jump at Fold 2 – system returns to resting state.

This mechanism explains the “**square-wave**” **bursting** pattern observed in biological experiments on glycolytic oscillations in yeast extracts



Conclusions & Future Directions



Key Findings

- ❑ The GL Model successfully translates a complex biological pathway into rigorous mathematics, capturing the essential nonlinear dynamics.
- ❑ Oscillations are not random fluctuations – they are governed by specific kinetic thresholds (Hopf bifurcation) arising from allosteric regulation.
- ❑ The fast-slow geometric analysis reveals relaxation oscillations as the underlying mechanism consistent with experimental observations.
- ❑ Demonstrates how cells utilize non-linear dynamics and feedback for metabolic efficiency and temporal organization.

References

- ❑ **Boiteux, A., Goldbeter, A., & Hess, B. (1975). Control of oscillating glycolysis of yeast by stochastic, periodic, and steady source of substrate: a model and experimental study. *Proceedings of the National Academy of Sciences*, 72, 3829–3833. <https://doi.org/10.1073/pnas.72.10.3829> Cited by: 308**
- ❑ **Goldbeter, A. (1974). Modulation of the adenylate energy charge by sustained metabolic oscillations. *FEBS Letters*, 43, 327–330. [https://doi.org/10.1016/0014-5793\(74\)80672-1](https://doi.org/10.1016/0014-5793(74)80672-1) Cited by: 56**
- ❑ **Monod, J., Wyman, J., & Changeux, J.-P. (1965). On the nature of allosteric transitions: A plausible model. *Journal of Molecular Biology*, 12, 88–118. [https://doi.org/10.1016/s0022-2836\(65\)80285-6](https://doi.org/10.1016/s0022-2836(65)80285-6) Cited by: 10989**
- ❑ **Tyson, J. J., & Novak, B. (2020). A Dynamical Paradigm for Molecular Cell Biology. *Trends in Cell Biology*, 30, 504–515. <https://doi.org/10.1016/j.tcb.2020.04.002> Cited by: 72**



Thank You!

